gpeyt1-16 · Cyclic stress ratio

GEO PE · Long Calculation Problems I — Topic 4: Earthquake Engineering & Dynamic Loads

Problem

A bridge abutment site is underlain by loose to medium dense clean sand. Water table at 6 ft. γ = 112 pcf above and γsat = 125 pcf below. The design earthquake has Mw = 7.0 and a site-adjusted peak ground acceleration amax = 0.32g (PGA on rock was 0.24g). Evaluate the sand at 24 ft depth using the Seed–Idriss simplified procedure with the mean rd relation.

Most nearly, the CSR at 24 ft is:

A. 0.338
B. 0.196
C. 0.319
D. 0.239

Given equations / data

> CSR = 0.65 (amax/g)(σvo/σ′vo) rd; rd = 1.0 − 0.00233z for z < 30 ft (z in ft); γw = 62.4 pcf

Worked solution

1. σvo = 6(112) + 18(125) = 2,922 psf
2. u = 18(62.4) = 1,123 psf → σ′vo = 1,799 psf
3. rd = 1 − 0.00233(24) = 0.9441
4. CSR = 0.65 × 0.32 × (2,922/1,799) × 0.9441 = 0.319

Final answer

0.319

Why the other values are wrong

- **0.338** — omitted rd
- **0.196** — omitted σvo/σ′vo
- **0.239** — used rock PGA instead of site amax

References (Jayce's library)

- FHWA-NHI-11-032 (GEC 3) Eq. 6-16, §6.4 Simplified Procedure; rd relation (Liao & Whitman) in GEC 3

Original problem inspired by the problem type in: https://www.youtube.com/watch?v=Hg3MTAvoJWw, https://www.youtube.com/watch?v=f9PIvbPStrU (no creator values copied).

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gpeyt1-17 · FS against liquefaction

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Problem

A silty sand layer (FC = 12%) at 18 ft has a clean-sand-equivalent (N1)60cs = 13, giving CRR(M7.5) = 0.17 from the base curve. The design event is Mw = 6.75 with amax = 0.28g; the magnitude scaling factor for that event is MSF = 1.19. Kσ = 1.0. Water table 8 ft; γ = 115 pcf above, γsat = 122 pcf below. Use rd = 1 − 0.00233z.

Most nearly, the factor of safety against liquefaction is:

A. 0.69
B. 1.22
C. 1.16
D. 0.82

Given equations / data

> CSR = 0.65 (amax/g)(σvo/σ′vo) rd; FSL = (CRR7.5 · MSF · Kσ)/CSR

Worked solution

1. σvo = 8(115) + 10(122) = 2,140 psf; σ′vo = 1,516 psf
2. rd = 0.9581; CSR = 0.65×0.28×1.412×0.9581 = 0.246
3. FSL = 0.17×1.19×1.0/0.246 = 0.82 (<1 → liquefiable)

Final answer

0.82

Why the other values are wrong

- **0.69** — omitted MSF
- **1.22** — inverted FS (CSR/CRR)
- **1.16** — used total stress for σ′vo (ratio = 1)

References (Jayce's library)

- FHWA-NHI-11-032 (GEC 3) Ch. 6 Eq. 6-16 and FS definition, MSF discussion

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gpeyt1-18 · Mononobe–Okabe

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Problem

An 18 ft cantilever wall retains level, dry granular backfill (γ = 120 pcf, φ = 34°, c = 0). The back face is vertical and wall friction is neglected (δ = 0). Seismic design uses kh = 0.16, kv = 0. The wall has a 10 ft base and a 2 ft thick stem; passive resistance at the toe is ignored.

Most nearly, the total (static + seismic) active thrust PAE is:

A. 7,403 lb/ft
B. 5,496 lb/ft
C. 14,807 lb/ft
D. 6,219 lb/ft

Given equations / data

> Mononobe–Okabe with δ = 0, β = 0, i = 0: $\theta = \tan^{-1}[kh/(1-kv)]$; $KAE = \cos^2(\varphi-\theta)/\{\cos^2\theta\,[1+\sqrt{(\sin\varphi\,\sin(\varphi-\theta)/\cos\theta)}]^2\}$; $PAE = \tfrac{1}{2}\,KAE\,\gamma\,H^2\,(1-kv)$

Worked solution

1. $\theta = \tan^{-1}(0.16) = 9.09°$
2. $KAE = \cos^2(34-9.09)/\{\cos^2(9.09)[1+\sqrt{(\sin34\cdot\sin24.91/\cos9.09)}]^2\} = 0.381$
3. (static Ka = 0.283)
4. $PAE = \tfrac{1}{2}(0.381)(120)(18)^2 = 7{,}403$ lb/ft

Final answer

7,403 lb/ft

Why the other values are wrong

- **5,496 lb/ft** — used static Rankine Ka
- **14,807 lb/ft** — forgot the ½
- **6,219 lb/ft** — multiplied by (1 − kh) instead of (1 − kv)

References (Jayce's library)

- NCEES PE Civil Reference Handbook v2.1 §3.1.5 Pseudostatic Analysis and Earthquake Loads (Mononobe–Okabe), p. 79; GEC 3 §11.3.1

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gpeyt1-19 · Pseudostatic rock slope

GEO PE · Long Calculation Problems I — Topic 4: Earthquake Engineering & Dynamic Loads

Problem

A highway rock cut has a daylighting planar joint dipping α = 35° with length L = 60 ft along the joint per ft of cut. The sliding block weighs W = 180,000 lb/ft. Joint strength: c = 500 psf, φ = 30°. The slope is fully drained (no water in the tension crack or on the joint). For seismic screening use k = 0.15. The rock unit weight is 160 pcf and RQD is 72% (not needed).

Most nearly, the pseudostatic factor of safety is:

A. 1.12
B. 0.85
C. 0.92
D. 1.18

Given equations / data

> FS = [cL + (W cos α − kW sin α − PWR − PWJ sin α) tan φ]/(W sin α + kW cos α + PWJ cos α); PWR = PWJ = 0 when drained

Worked solution

1. cL = 500×60 = 30,000 lb/ft
2. Normal: W cos α − kW sin α = 147,447 − 15,487 = 131,961 lb/ft
3. Resisting = 30,000 + 131,961×tan30 = 106,188
4. Driving = W sin α + kW cos α = 125,361
5. FS = 0.85 (static 1.12)

Final answer

0.85

Why the other values are wrong

- **1.12** — static FS (dropped k)
- **0.92** — added k only to the driving side
- **1.18** — inverted FS

References (Jayce's library)

- NCEES PE Civil Reference Handbook v2.1 §3.6.3 Rock Slope Failure (Stability of Rock Slope figure), p. 104

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gpeyt1-20 · Pseudostatic infinite slope

GEO PE · Long Calculation Problems I — Topic 4: Earthquake Engineering & Dynamic Loads

Problem

A long, dry, cohesionless fill slope is inclined at β = 24° (about 2.25H:1V) with φ′ = 36°, γ = 120 pcf. A shallow failure surface parallel to the slope at 6 ft depth is being checked. The seismic coefficient is kh = 0.12 applied horizontally (out of slope) at the block's center of mass. No pore pressure.

Most nearly, the pseudostatic factor of safety is:

A. 1.63
B. 1.46
C. 1.35
D. 1.22

Given equations / data

> Static infinite slope (ru = 0): F = tan φ′/tan β. Resolve W and khW parallel/normal to the slip plane: F = (cos β − kh sin β) tan φ′/(sin β + kh cos β)

Worked solution

1. Static F = tan36/tan24 = 1.63
2. Normal (per W): cos24 − 0.12 sin24 = 0.8647
3. Driving (per W): sin24 + 0.12 cos24 = 0.5164
4. F = 0.8647×tan36/0.5164 = 1.22 (depth and γ cancel)

Final answer

1.22

Why the other values are wrong

- **1.63** — static FS only
- **1.46** — divided static FS by (1 + kh)
- **1.35** — added kh·sin β to the normal force (wrong sign)

References (Jayce's library)

- NCEES PE Civil Reference Handbook v2.1 §3.6.4 Infinite Slope, p. 105; pseudostatic resolution as in §3.6.3 (kW terms); GEC 3 Ch. 7

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